says Dr Dattatray Solanke, gastroenterologist, Kokilaben Dhirubhai Ambani Hospital, Mumbai. Prompted by the ENS, the gut acts in anticipation of food, inducing salivation and gastric secretion. Inside the gut, the microbiota crank into action. They digest bile acids from the liver and amino acids to produce chemicals that assist the brain in releasing chemical messengers called neuropeptides, among them Glanin, which influences our sleep, mood and blood pressure.

Microbiota in the gut also produce short-chain fatty acids (SCFAs) such as butyrate, propionate and acetate when digesting fibre in food. SCFAs affect brain function, immune regulation and interact with nerve cells in our autonomic nervous system, which dictates our flight or fight response. "The human gut and brain interact, impacting each other's functionality. Specific bacteria have the capability of secreting molecules that affect the brain-gut axis," says Dr Vikas Singla, senior director & head of gastroenterology, Max Superspeciality Hospital in Delhi.

Human microbiota are also said to exert an influence in the release of powerful neurotransmitters that are vital for our day-to-day and long-term well-being. Among these are GABA, or gamma-aminobutyric acid, which helps control fear and anxiety; and dopamine, which is associated with pleasure and reward. Ninety-five per cent of the hormone Serotonin, which acts as a neurotransmitter that aids digestion, boosts mood, controls sleep, clots blood and influences learning and memory, is also produced in the gut. "That the gut influences emotional response is visible when we consider anxiety or psychogenic vomiting, triggered by physical stress. Again, people with depression don't feel like eating, or at times stress or emotional instability end up in a 'binge eating disorder'," says Dr Kanishka Meel, a psychiatrist who consults with telehealth platform Practo.

Perils of an Unhealthy Gut

"A healthy gut can enhance cognitive flexibility, attention span and emotional resilience," says Dr Harsh Seth, consultant laparoscopic gastrointestinal surgeon at Wockhardt Hospitals, Mumbai. But a troubled one can have implications beyond the gut. "Poor gut health can lead to increased systemic inflammation, which has been associated with cognitive decline and mood disorders," says Dr Lohith U., consultant, surgical gastroenterology, bariatric and GI oncology, Manipal Hospitals, Bengaluru.

The 2023 Gut Health Survey found that nearly 6 in 10 people who have poor gut health have mental health issues like mood swings and anxiety. "Earlier, it was thought that mental health problems lead to issues in the gut, but now it is established that some mental disorders start with problems in the gut," says Dr Purnima Nagaraja, a mental health professional from Hyderabad.

The gut microbiome also plays a significant role in immune system regulation. A 2023 study in the journal *Gut Pathogens* revealed how dysbiosis impacted the severity of Covid-19, with gut microbiota influencing the release of cytokines. Inflammation in the body, in turn, can lead to a leaky gut, when toxins and bacteria from the gut enter the bloodstream.

Gut issues, in fact, were found to be associated with 20 per cent of all cancers, according to a 2022 study in the journal *Science Direct*. They may also affect our ability to socialise. A 2014 study in the journal *Psychoneuroendocrinology* found that rats raised in sterile conditions were less sociable than those with intact gut microbiota. Studies are also trying to ascertain exactly which gut bacteria impact what bodily function. It should help doctors effectively treat a whole array of diseases.



"Probiotics, prebiotics, fruits and vegetables could protect against mental disorders by enhancing beneficial gut microbiota"

—DR VIKAS SINGLA
Senior Director & Head of Gastroenterology, Max Superspeciality Hospital, Delhi

MIND YOUR GUT

Reducing excess sugar and fat does wonders for your gut. You should also drink enough water to keep yourself hydrated. To cultivate a healthy microbiome, change your diet to include prebiotics or dietary fibre through complex carbs, fruit and vegetables, and probiotics, or fermented foods. A quick guide



KALE, COLLARD GREENS, SPINACH

These leafy greens are high in fibre, and also have an unusual form of sugar called sulfoqui-